Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 3

Question 1

Which of the following statement is false?

- $\log(f(n)g(n)) = O(\max(\log f(n), \log g(n)))$
- $2^{n+1} = O(2^n)$
- $(n+1)! = O(n!)$
- $(n+a)^b = \Theta(n^b)$

Question 1

Answer: C

$$(n+1)! = (n+1) \times n!$$

is not $O(n!)$ as

$$\frac{(n+1)!}{n!} = n+1 > C$$

for any constant C

Question 1

Other answers are correct:

A. $\log f(n)g(n) = \log f(n) + \log g(n)$

$$= O(\max\{\log f(n), \log g(n)\}$$

B. $2^{n+1} = 2 \times 2^n = O(2^n)$

D. $(n+a)^b = \Theta(n^b)$

as $(n+a)^b$ is a polynomial of degree *b*

Question 2

Use the master method to give tight asymptotic bound for $T(n) = 2T(n/4) + 1$.

- $\Theta(\sqrt{n})$
- $\Theta(n)$
- $\Theta(\log n)$
- $\Theta(n \log n)$

Question 2

Answer: A

$T(n) = 2T(n/4) + 1$

Master Theorem case 1

$$f(n) = 1 = O(n^{\log_4 2 - \epsilon}) = O(n^{1/2 - \epsilon})$$

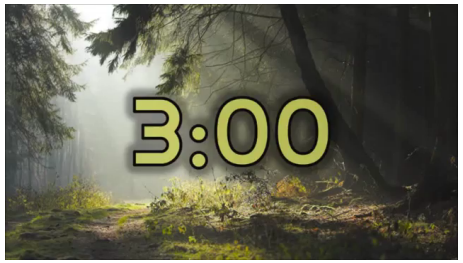
Hence

$$T(n) = \Theta(\sqrt{n})$$

Question 3

Use the master method to give tight asymptotic bound for $T(n) = 2T(n/4) + n$.

- $\Theta(\log n)$
- $\Theta(\sqrt{n})$
- $\Theta(n)$
- $\Theta(\sqrt{n} \log n)$



Question 3

Answer: C

$T(n) = 2T(n/4) + n$

Master Theorem case 3

$$f(n) = n = \Omega(n^{\log_b a+\epsilon}) = \Omega(n^{1/2+\epsilon})$$

and

$$af(n/b) = 2f(n/4) = 2n/4 = n/2 \leq f(n) = n$$

Hence $T(n) = O(n)$

Question 4

Use the master method to give tight asymptotic bound for $T(n) = 4T(n/2) + n^2 \log n$.

- $\Theta(n \log n)$
- $\Theta(n^2 \log n)$
- $\Theta(n^2 \log^2 n)$
- $\Theta(n^2)$

Question 4

Answer: C

$T(n) = 4T(n/2) + n^2 \log n$

Master Theorem case 2

$$f(n) = n^2 \log n = \Theta(n^{\log_2 4} \log n)$$

Hence

$$T(n) = \Theta(n^2 \log^2 n)$$